

Random walk

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The random walk can be mathematically modeled using random variables. Let an idealized point particle travelling in two dimensions be a distance X_n from the origin after n time steps. Every time step, we add a vector $(S \cos(\theta_i), S \sin(\theta_i))$ to its position vector, which starts at $(0, 0)$. Here, S is the fixed distance travelled every step and θ_i is a random angle between the horizontal axis and the direction of motion at the i th step. Our aim is to find the expectation of X_n . X_n can be computed by finding the magnitude of the position vector:

$$\begin{aligned} X_n &= |(S \cos(\theta_1), S \sin(\theta_1)) + (S \cos(\theta_2), S \sin(\theta_2)) + \dots + (S \cos(\theta_n), S \sin(\theta_n))| \\ &= S \sqrt{(\cos(\theta_1) + \dots + \cos(\theta_n))^2 + (\sin(\theta_1) + \dots + \sin(\theta_n))^2} \end{aligned}$$

Note that graphically $E(\cos(\theta_i))$ and $E(\sin(\theta_i))$ are both zero since they have zero signed area underneath them from 0 to 2π .

$$E(\cos^2(\theta_i)) = 1/2\pi \int_0^{2\pi} \cos^2(\theta_i) d\theta_i = 1/4\pi \int_0^{2\pi} \cos(2\theta_i) + 1 d\theta_i = 1/2$$

$$E(\sin^2(\theta_i)) = 1/2\pi \int_0^{2\pi} \sin^2(\theta_i) d\theta_i = 1/4\pi \int_0^{2\pi} -\cos(2\theta_i) + 1 d\theta_i = 1/2$$

The square of the sum in the expression for X_n equals a sum of the squares of cosines and sines assuming that the random angle at each step is independent of the random angles at other steps. This gives:

$$E(X_n^2) = E(S^2((\cos(\theta_1) + \dots + \cos(\theta_n))^2 + (\sin(\theta_1) + \dots + \sin(\theta_n))^2))$$

$$\begin{aligned}
E(X_n^2) &= E(S^2((\cos(\theta_1) + \dots + \cos(\theta_n))^2 + (\sin(\theta_1) + \dots + \sin(\theta_n))^2)) \\
E(X_n^2) &= E(S^2 \sqrt{(\cos(\theta_1) + \dots + \cos(\theta_n))^2 + (\sin(\theta_1) + \dots + \sin(\theta_n))^2}) \\
&= E(S^2 \cos^2(\theta_1) + \dots + \cos^2(\theta_n) + \sin^2(\theta_1) + \dots + \sin^2(\theta_n)) \\
&= S^2(1/2 + 1/2 \dots + 1/2) = S^2 n
\end{aligned}$$

Thus we obtain the root mean square of the distance travelled after n steps of length S as $S\sqrt{n}$